

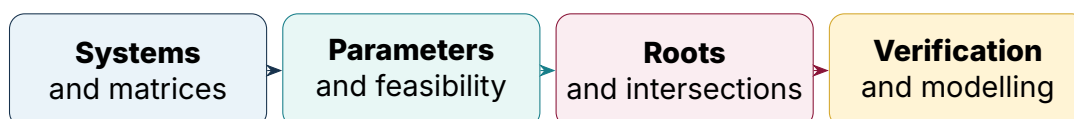
1.6

Technology for Equations and Systems

Full Notes - Linear Systems - Polynomial Roots - Intersections - Verification - TI-84

Central idea. Technology can produce a candidate solution. Mathematics decides whether the output is **complete**, **verified**, and **admissible**.

Model → Solve → Verify → Interpret



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Class: _____

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Version: Student Full Notes

Learning route and technology evidence map

model → transparent entry → complete output → independent check

Learning intentions

By the end of Lesson 1.6, you should be able to:

1. solve and classify two-variable linear systems;
2. solve three-variable systems while preserving variable and coefficient order;
3. determine unknown model parameters from independent conditions;
4. find all relevant real polynomial roots and identify multiplicity;
5. solve exact and numerical intersections using algebra, graphs, and TI-84 tools appropriately;
6. verify calculator output with substitution, residuals, factorization, or a second representation;
7. apply domain, units, integrality, non-negativity, precision, and capacity restrictions;
8. communicate a reproducible technology method and a defensible contextual conclusion.

Core reference sheet

2-variable system

A solution (x, y) must satisfy *both* original equations.

3-variable system

Use the same variable order in every row; missing variables require coefficient 0.

Augmented matrix

$[A \mid b]$ stores coefficients and constants in one matrix.

RREF

$\text{rref}([A \mid b])$ exposes system solutions and can reveal inconsistency/dependence.

Root/zero/x-intercept

$p(r) = 0 \iff r$ is a root $\iff (r, 0)$ is an x-intercept.

Intersection

$f(x) = g(x) \iff f(x) - g(x) = 0$.

Residual

$r_i = \text{LHS}_i - \text{RHS}_i$; exact solutions give $r_i = 0$.

Multiplicity

Even multiplicity touches/turns; odd multiplicity crosses; higher multiplicity is flatter near the root.

Admissibility

Apply domain, units, integrality, non-negativity, and capacity after solving.

Technology principle

GDC EXPECTED This lesson explicitly trains the TI-84 Plus CE because solving systems, locating polynomial zeros, and finding graph intersections are genuine technology skills. **GDC SKILL** Record the equations, variable order, matrix dimensions, graph window or bounds, complete output, verification, and interpretation. **GDC NOT NEEDED** Classification, exact factorization, and obvious algebraic structure should still be recognized without hiding behind a calculator.

IB communication tip

A statement such as “my calculator gave $(4, 3)$ ” is incomplete. A strong solution records *what was entered*, preserves the variable order, reports the complete output, checks the original conditions, and states why the result is admissible.

1.6A - Two-Variable Systems and Classification

common solutions and transparent matrix evidence

What a system solution means

For

$$\begin{cases} a_1x + b_1y = c_1, \\ a_2x + b_2y = c_2, \end{cases}$$

a solution (x, y) makes **both** equations true. Graphically, it is a common point of both graphs.

Three system classes

Unique	one common point	two independent conditions determine one ordered pair
Inconsistent	no common point	the conditions contradict each other
Dependent	infinitely many common points	one equation repeats the information in the other

Worked Example 1**GDC EXPECTED**

Solve

$$\begin{cases} 2x + y = 11, \\ x - y = 1. \end{cases}$$

Record the variable order, solve, and verify both equations.

WORKING / ANNOTATIONS**Your Turn 1****GDC EXPECTED**

Solve and verify

$$\begin{cases} 3x + 2y = 16, \\ x + y = 6. \end{cases}$$

Use the order x, y .**WORKING / ANNOTATIONS****Matrix form for a 2x2 system**

The augmented matrix for

$$2x + y = 11, \quad x - y = 1$$

is

$$\left[\begin{array}{cc|c} 2 & 1 & 11 \\ 1 & -1 & 1 \end{array} \right].$$

The third column is not a third variable; it stores constants.

Worked Example 2

GDC NOT NEEDED

Classify each system.

$$(a) \begin{cases} 2x + 4y = 8 \\ x + 2y = 4 \end{cases} \quad (b) \begin{cases} 2x + 4y = 8 \\ x + 2y = 5 \end{cases}$$

WORKING / ANNOTATIONS

Your Turn 2

GDC NOT NEEDED

Classify each as unique, inconsistent, or dependent.

1. $y = -3x + 4$ and $y = x - 2$;
2. $3x - 6y = 9$ and $x - 2y = 3$;
3. $4x + 2y = 7$ and $2x + y = 5$.

WORKING / ANNOTATIONS

Common trap

Checking only one equation is not verification of a system. A candidate must satisfy every original equation.

Checkpoint - Two-variable systems

1. Solve $5x - 2y = 4$ and $3x + y = 13$ exactly.
2. Classify $6x + 3y = 12$ and $2x + y = 4$.
3. Explain what a graph intersection represents algebraically.
4. Write the augmented matrix for $4x - 3y = 7$ and $2x + y = 9$.

WORKING / ANNOTATIONS

1.6B - Three Variables, Parameter Fitting, and Feasibility

coefficient order is part of the mathematics

Protect the variable order

For a 3-variable solver, choose an order such as x, y, z and keep it in every row. If a variable is missing, enter a zero coefficient:

$$3x + 2z = 17 \implies [3, 0, 2 \mid 17].$$

A plausible answer from a scrambled coefficient order is still wrong.

Worked Example 3

GDC EXPECTED

A school event sells 300 adult, student, and child tickets. Adult tickets cost QAR 40, student tickets QAR 25, and child tickets QAR 15. Total revenue is QAR 8700, and $s = 2c$. Find a, s, c .

WORKING / ANNOTATIONS

Your Turn 3

GDC EXPECTED

Solve

$$\begin{cases} x + y + z = 9, \\ 2x - y + z = 4, \\ x + 2y - z = 3. \end{cases}$$

State the order used and verify the triple.

WORKING / ANNOTATIONS

Parameter fitting is a system problem

Unknown model parameters can be treated as system variables. For a line $L(x) = mx + c$ passing through $(2, 11)$ and $(5, 23)$:

$$2m + c = 11, \quad 5m + c = 23.$$

Solving gives $m = 4$, $c = 3$, so $L(x) = 4x + 3$.

Worked Example 4

GDC EXPECTED

A calibration curve is $P(t) = at^2 + bt + c$ with

$$P(0) = 4, \quad P(2) = 18, \quad P(-1) = 3.$$

Determine a, b, c .

WORKING / ANNOTATIONS

Your Turn 4

GDC NOT NEEDED

Using the calibrated model $P(t) = 2t^2 + 3t + 4$, solve $P(t) = 39$. If t is elapsed time, identify the relevant value.

WORKING / ANNOTATIONS

Feasibility checklist

A solver output must also satisfy the context:

- non-negative quantities when negative values have no meaning;
- integer values when variables count objects;
- stated capacity or range limits;
- units and domain restrictions.

A result such as $(45, 38, -3)$ for item counts can satisfy the equations yet still be infeasible.

Your Turn 5

GDC NOT NEEDED

A resource-allocation solver returns $(x, y, z) = (45, 38, -3)$, where all variables count items. Explain what the result establishes and what should happen next.

WORKING / ANNOTATIONS

Checkpoint - Three variables and parameters

1. Explain why $3x + 2z = 17$ must be entered as coefficients 3, 0, 2 in an x, y, z solver.
2. A line passes through (1, 7) and (4, 19). Find its equation.
3. State two contextual restrictions that can reject an algebraically correct system solution.
4. For $Q(t) = at^2 + bt + c$, write the three equations generated by $Q(0) = 2$, $Q(1) = 6$, $Q(3) = 20$.

WORKING / ANNOTATIONS**1.6C - Polynomial Roots, Intersections, and Numerical Solving**

completeness and field of view matter

Root, zero, and x-intercept

$$p(r) = 0 \iff r \text{ is a root} \iff (r, 0) \text{ is an x-intercept}$$

A polynomial root tool solves $p(x) = 0$. A graph locates where $y = p(x)$ has y-coordinate zero.

Degree and multiplicity

A degree- n polynomial has n complex roots counted with multiplicity. The number of *distinct real* roots can be smaller.

- multiplicity 1: the graph crosses the x-axis;
- multiplicity 2: the graph touches and turns;
- multiplicity 3: the graph crosses but flattens near the root.

Worked Example 5

GDC EXPECTED

Find all real roots of

$$x^3 - 4x^2 - x + 4 = 0.$$

WORKING / ANNOTATIONS

Your Turn 6

GDC EXPECTED

Find all real roots of $x^3 - 3x^2 - 4x + 12 = 0$ and verify algebraically.

WORKING / ANNOTATIONS

Worked Example 6

GDC EXPECTED

Solve $x^3 - 3x^2 + 4 = 0$ and identify multiplicity.

WORKING / ANNOTATIONS

Your Turn 7

GDC NOT NEEDED

A degree-4 solver lists $-3, 1, 1, 5$. State the distinct real roots, multiplicity of each, and the number of x-intercepts.

WORKING / ANNOTATIONS

Intersections are equation solutions

To solve $f(x) = g(x)$, either graph both functions or solve one zero equation:

$$f(x) = g(x) \iff f(x) - g(x) = 0.$$

If the question asks for *intersection points*, report both coordinates.

Worked Example 7

GDC EXPECTED

Find the intersections of

$$y = 2x + 1 \quad \text{and} \quad y = x^2 - 3.$$

WORKING / ANNOTATIONS

Your Turn 8

GDC EXPECTED

Solve

$$5x + 12 = 80(0.9)^x, \quad x \geq 0.$$

Record the two functions, a suitable window, the solution, and a substitution check.

WORKING / ANNOTATIONS

A graph has a field of view

One graph screen cannot prove there are no other solutions. Protect against hidden roots or intersections by:

- stating the domain first;
- scanning with a broad graph or table before zooming;
- widening the window at least once when completeness matters;
- using more than one initial guess if a numerical solver can converge to different roots;
- remembering that even-multiplicity roots may not change sign.

Worked Example 8

GDC EXPECTED

For

$$h(t) = -4.9t^2 + 18t + 1.5,$$

find the times when $h = 0$ and identify the physically admissible elapsed time.

WORKING / ANNOTATIONS

Checkpoint - Roots and intersections

1. Find all real roots of $x^4 - 10x^2 + 9 = 0$.
2. Explain why a repeated root can be missed by a sign-change scan.
3. Find both intersections of $y = 2x + 1$ and $y = x^2 - 3$.
4. State two actions that reduce the risk of a hidden graph intersection.

WORKING / ANNOTATIONS

1.6D - Verification, Constraints, Modelling, and Mastery

a calculator result is the beginning of the conclusion

Residuals

For equation i ,

$$r_i = \text{LHS}_i - \text{RHS}_i$$

Exact solutions give residual 0. Rounded outputs should produce residuals small relative to the reporting precision and the scale of the problem.

Worked Example 9

GDC EXPECTED

A solver gives $(x, y) = (2.727273, 4.818182)$ for

$$5x - 2y = 4, \quad 3x + y = 13.$$

Verify the rounded output and identify the exact solution.

WORKING / ANNOTATIONS

Your Turn 9

GDC EXPECTED

A candidate solution produces residuals 2.0×10^{-5} and -1.0×10^{-5} in a model reported to three decimal places. Find the maximum absolute residual and comment.

WORKING / ANNOTATIONS

Common trap

A small residual verifies the equations that were entered. It cannot detect a modelling mistake if the wrong equations were entered perfectly. Compare the equations with the original wording or data.

Guard digits and final rounding

Keep unrounded calculator values during substitution and later calculations. Round only once at the final reporting stage unless the context requires an exact value.

Worked Example 10**GDC EXPECTED**

Revenue is $R(x) = 52x - 0.04x^2$ and cost is $C(x) = 900 + 18x$, where $0 \leq x \leq 700$. Find the break-even quantities and determine which are admissible.

WORKING / ANNOTATIONS**Your Turn 10****GDC EXPECTED**

For the same model, determine the number of units that maximizes profit and the maximum profit.

WORKING / ANNOTATIONS**Continuous output versus discrete reporting**

If technology gives a crossing at $n = 18.24$ but n counts completed whole periods, do not report 18.24 periods. Check periods 18 and 19 and report the first admissible period satisfying the original condition.

Worked Example 11

GDC EXPECTED

Two populations are modelled by

$$A(t) = 500(1.12)^t, \quad B(t) = 900(1.04)^t.$$

Find when they are equal and the first whole year for which $A > B$.

WORKING / ANNOTATIONS

Checkpoint - Verification and modelling

1. Define a residual and explain what a small residual does and does not prove.
2. A continuous threshold occurs at $n = 12.23$ days and measurements are made at whole days. State the first recorded day after the crossing.
3. Explain why the second break-even root 822.6493 is rejected in a model restricted to $0 \leq x \leq 700$.
4. State three items that should appear in a reproducible TI-84 solution.

WORKING / ANNOTATIONS

TI-84 Classroom Training - Six Required Workflows

corrected official paths used by the live lesson

TI-84 route policy

The primary system workflow is the reproducible **MATRIX** → **EDIT** → **rref()** route. If PlySmlt2 is installed on a physical TI-84 Plus CE, it can be used as a second check. For roots and intersections, the graph commands are part of the method because they require bounds, windows, and completeness decisions.

A1 - 2x2 system with RREF

GDC SKILL

Problem: $2x + y = 11$, $x - y = 1$.

1. Press $2\text{nd} \rightarrow \text{x}^{-1}$ (MATRIX).
2. Move to EDIT and choose 1: [A].
3. Set dimensions to 2×3 .
4. Enter rows $[2, 1 \mid 11]$ and $[1, -1 \mid 1]$. Use the physical $(-)$ key for the negative coefficient.
5. Press $2\text{nd} \rightarrow \text{MODE}$ (QUIT).
6. Open MATRIX $\rightarrow$ MATH, choose $\text{rref}()$, paste [A], close the bracket, and press ENTER.

$$\text{rref}([A]) = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 3 \end{bmatrix}.$$

Verify: $2(4) + 3 = 11$, $4 - 3 = 1$.

B1 - 3x3 contextual system with RREF

GDC SKILL

Problem: ticket model with variables a, s, c .

1. MATRIX $\rightarrow$ EDIT $\rightarrow$ [A]; set 3×4 .
2. Enter $[1, 1, 1 \mid 300]$.
3. Enter $[40, 25, 15 \mid 8700]$.
4. Enter $[0, 1, -2 \mid 0]$ for $s = 2c$.
5. QUIT, then evaluate $\text{rref}([A])$.

$$a = 120, \quad s = 120, \quad c = 60.$$

Verify all three conditions before interpreting the ticket counts.

C1 - Cubic roots with Zero

GDC SKILL

Problem: $x^3 - 4x^2 - x + 4 = 0$.

1. Press Y= and enter $Y_1 = x^3 - 4x^2 - x + 4$.
2. Use ZOOM $\rightarrow$ 6:ZStandard, then inspect whether all relevant intercepts are visible.
3. Press $2\text{nd} \rightarrow \text{TRACE}$ (CALC) $\rightarrow$ 2:zero.
4. Set a left bound, right bound, and guess around one intercept.
5. Repeat Zero separately around each remaining intercept.

$$x = -1, 1, 4.$$

Audit completeness: a cubic has three roots counted with multiplicity; factorization verifies the full real list here.

C2 - Exact intersections with Intersect

GDC SKILL

Problem: $y = 2x + 1$ and $y = x^2 - 3$.

1. Press $Y=$ and enter both functions.
2. Graph and confirm that both crossings are visible.
3. Press $2^{nd} \rightarrow TRACE \text{ (CALC)} \rightarrow 5:intersect$.
4. Press ENTER for First curve and Second curve, then move near the desired crossing and press ENTER for Guess.
5. Repeat near the second crossing.

$$(-1.236068, -1.472136), \quad (3.236068, 7.472136).$$

Algebra supplies the exact coordinates $(1 \pm \sqrt{5}, 3 \pm 2\sqrt{5})$.

C3 - Numerical model intersection

GDC SKILL

Problem: $5x + 12 = 80(0.9)^x, x \geq 0$.

1. Set $Y_1 = 5x + 12, Y_2 = 80(0.9)^x$.
2. Choose a contextual WINDOW, for example $0 \leq x \leq 15$, with a y-range containing both graphs.
3. GRAPH and inspect the number of relevant crossings; widen the window once if completeness matters.
4. Use $CALC \rightarrow 5:intersect$, select both curves, and supply a guess near the crossing.

$$x \approx 6.05443, \quad y \approx 42.2722.$$

Verify: both original expressions evaluate to approximately 42.2722.

D1 - Rounded RREF output and residuals

GDC SKILL

Problem: $5x - 2y = 4, 3x + y = 13$.

1. Enter the 2×3 augmented matrix $[5, -2 | 4], [3, 1 | 13]$.
2. Evaluate $rref([A])$.
3. Retain exact fractions if displayed: $x = 30/11, y = 53/11$.
4. If decimals are needed, keep guard digits: $x \approx 2.727273, y \approx 4.818182$.
5. Evaluate both residuals using the unrounded stored values where possible.

Conclusion: both residuals are negligible at six-decimal precision.

IB-style Mixed Practice

model - solve - verify - interpret

Your Turn 11

GDC EXPECTED

A rectangle has area 84 cm^2 and its length is 5 cm more than its width. Use technology if helpful to find the physically valid width.

WORKING / ANNOTATIONS

Your Turn 12

GDC EXPECTED

A line $L(x) = mx + c$ passes through $(2, 11)$ and $(5, 23)$. Determine m, c and verify both points.

WORKING / ANNOTATIONS

Your Turn 13

GDC EXPECTED

A degree-4 solver reports roots $-2, 3, 3, 6$. How many distinct x-intercepts are represented, and what does the repeated 3 mean?

WORKING / ANNOTATIONS

Your Turn 14

GDC EXPECTED

For $h(t) = -4.9t^2 + 16t + 1$, technology gives two real roots. Determine the positive root to three decimal places and explain the rejection of the other root.

WORKING / ANNOTATIONS

Your Turn 15

GDC EXPECTED

A production model gives algebraic roots -6 and 14 , where the variable counts units. State the admissible root and justify.

WORKING / ANNOTATIONS

IB task checklist

Before submitting any technology-based response, check that you have:

- defined variables and restrictions;
- written the equations or functions before using technology;
- recorded variable order, matrix dimensions, graph window, bounds, or initial guess when relevant;
- reported the complete solution set or all relevant intersections;
- verified the output independently;
- rejected inadmissible values for a stated mathematical reason;
- included units and appropriate rounding in the final sentence.

Final Mastery Check

can you defend every calculator answer?

Checkpoint - Mixed mastery

1. Solve and verify the system $2x + 3y = 13$, $4x - y = 5$.
2. Solve $x^3 - 2x^2 - 5x + 6 = 0$ and state the complete real root set.
3. A quadratic model gives roots -12 and 7 for a physical length. State the admissible answer and explain why.
4. Explain the difference between a repeated root and a repeated x-intercept.
5. State the official TI-84 routes for (a) Zero and (b) Intersect.
6. Explain why a perfectly small residual can still accompany a wrong contextual model.

WORKING / ANNOTATIONS**Mastery reflection**

I can now:

- solve 2x2 and 3x3 systems with a transparent matrix workflow;
- distinguish unique, inconsistent, and dependent systems;
- fit unknown parameters from data conditions;
- find complete polynomial root sets and interpret multiplicity;
- locate exact and numerical intersections without hiding relevant solutions;
- use residuals, domain restrictions, units, and contextual constraints to validate output;
- reproduce the six Lesson 1.6 TI-84 workflows and explain why each result is trustworthy.